

Mourad Kadri

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EXPERIENCE

- **Banque de Tunisie et des Émirats (BTE), Tunisia** *Feb 2026 – Present*
 - Designed and validated the NLU model layer for an automated voice agent: fine-tuned a Rasa 3.6 classifier achieving 98.1% F1 across 39 intents on a 3,704-example test set, with rigorous train/test splitting and error analysis.
 - Built and evaluated a retrieval pipeline (ChromaDB + bge-m3 embeddings) over policy documents, benchmarking retrieval latency and relevance against an LLM-fallback baseline (Cerebras).
 - Fine-tuned an int8 quantized faster-Whisper model for domain-specific ASR and a DistilBERT classifier for multilingual distress/intent detection, optimizing for both accuracy and inference latency on constrained hardware.
 - Designed a 65-case adversarial test suite and 69 dialogue story tests to statistically validate model robustness before deployment, tracking 135/138 deployment validation checks.
- **MyAI Pathways, Remote (USA)** *Oct 2025 – Dec 2025*
 - Developing an AI model readiness assessment tool for businesses using LangChain and Next.js, enabling companies to evaluate AI integration potential.
- **Inetum, Tunisia** *Jul 2025 – Sept 2025*
 - Built an AI agent for Sage X3 support leveraging SQL query generation and CSV data manipulation for automated data extraction and reporting.
 - Developed a TypeScript-based mobile solution for Sage X3 to enable seamless ERP access.
- **MEDIANET, Tunisia** *Feb 2023 – Jun 2023*
 - Designed a user analytics dashboard using Chart.js and MongoDB aggregation pipelines for behavioral and engagement metrics.
 - Led development of a MERN & Flutter-based reward system with real-time notifications and gamification.
 - Established a CI/CD pipeline with GitHub Actions, reducing deployment time by 90%.

KEY PROJECTS

- **Emotion Recognition:** Built an audio classification pipeline with MFCC feature extraction and LSTM models for emotion detection in voice data.
- **RAG AI Resume Matcher & Skill Coach:** Designed a resume-matching and skill-grading model with a course recommendation engine using LangChain.
- **Exaq OCR Assistant:** Built a document classification/extraction pipeline (YOLOv13 + TrOCR) achieving 97% accuracy on mixed handwritten/printed text, reducing data entry time by 85%.
- **MLOps Pipeline:** Implemented experiment tracking with MLflow, model versioning via DVC, and reproducible containerized training/deployment with Docker.
- **LABLAB.AI Hackathon: Language Learning Platform:** Developed a GPT-4-powered platform using Flask, MongoDB, and Docker for language proficiency.

EDUCATION

- **ESPRIT, Tunisia** – Data Science Engineering (Final Year) *2023 – 2026*
- **ISTIC, Tunisia** – Research MSc in Robotics *2023 – 2025*
- **ISTIC, Tunisia** – BSc Software Engineering *2022 – 2023*
- **Lycée Ibnou Rachik** – Baccalaureate in Technical Sciences *2020*

SKILLS & CERTIFICATIONS

Skills:

Machine Learning/AI: TensorFlow, Scikit-learn, YOLO, HuggingFace Transformers, MLflow, FAISS, LangChain, Rasa, Whisper, ChromaDB

Programming Languages: Python, SQL, JavaScript (Node.js), Bash

Databases: MongoDB, MySQL, SQLite

DevOps: Docker, GitHub Actions, DVC, CI/CD

Certifications:

Cisco AI Fundamentals, NVIDIA LLMs, Datacamp ML/AI Stack

Languages:

Arabic, French, English